

Module 1

Module-1 (5 Hours)

Introduction: Meaning of Research, Objectives of Engineering Research, and Motivation in Engineering Research, Types of Engineering Research, Finding and Solving a Worthwhile Problem.

Ethics in Engineering Research, Ethics in Engineering Research Practice, Types of Research Misconduct, Ethical Issues Related to Authorship.

Meaning of Research



Research

Refers to “**Search for knowledge**”.

It is a **scientific and systematic search** for particular information on a specific topic.

It is an **art of scientific investigation**.

It is a careful **investigation or enquiry** specially through **search for new facts** in any branch of knowledge.

It is a **movement from known to the unknown**.

It is usually the **voyage (a long journey) of discovery**

- *We all possess the vital instinct of inquisitiveness.* This inquisitiveness is the mother of all knowledge and the method, which one employs for obtaining the knowledge of whatever the unknown, can be termed as **research**.
- Clifford woody -> Research comprises of
 - defining and redefining the problems,
 - formulating hypothesis or suggested solution;
 - collecting, organising and evaluating data;
 - making deductions and reaching conclusions;
 - at last carefully testing the conclusions to determine whether they fit the formulating hypothesis.

Example:-

Defining the problem

- “**The current battery used in electric cars takes too long to charge**”.

Formulating a Hypothesis (Suggested Solution)

- A possible solution can be “**Using a different material (say, graphene) in the battery could reduce charging time.**”

Collecting, organizing and evaluating data

- To test this, the engineering team **collects data on different materials' properties, conducts experiments with graphene-based batteries,** and **records the charging time for each test.**

Making deductions and reaching conclusions

- After **analyzing the data**, the team may conclude: “**Graphene-based batteries charge 50% faster than current batteries.**”

Testing the conclusion

- The final step is to test this conclusion by **repeating the experiment in different conditions (temperature, usage) to ensure that the new material (graphene) consistently improves charging time.**

- D.Slesinger and M.Stephenson in the encyclopedia of social sciences define research as “**the manipulation of things, concepts or symbols for the purpose of generalizing to extend, correct or verify knowledge**, whether that knowledge aids in construction of theory or in the practice of an art”.

Example: Software testing

Manipulation

- Testers create various test cases, manipulating inputs, data, and environmental conditions.

Purpose

- The goal is to identify defects, assess the software's functionality, and ensure it meets specified requirements.

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Knowledge extension

- By executing these tests, developers gain insights into the software's behavior, potential issues, and areas for improvement. This knowledge helps them refine the code, enhance performance, and build more robust and reliable software.

- Research is, thus, an **original contribution to the existing stock of knowledge making for its advancement.**
- The **search for knowledge through objective and systematic method of finding solution to a problem** is **research.**
- The term 'research' refers to the **systematic method consisting of enunciating the problem, formulating a hypothesis, collecting the facts or data, analysing the facts and reaching certain conclusions either in the form of solution(s) towards the concerned problem or in certain generalizations for some theoretical formulation.**

Example: Developing a Water Purification System

Enunciating the Problem:

Imagine a **village has a problem with contaminated drinking water, leading to health issues**. The problem statement could be: **"How can we develop an affordable water purification system for the village?"**



Formulating a Hypothesis:

A possible solution (hypothesis) could be: **"Using charcoal and sand filters can effectively remove contaminants from the water, making it safe for drinking."**



Collecting the Facts or Data:

The researcher gathers data by **studying various water filtration methods, testing the quality of the contaminated water, and running experiments with charcoal and sand filters**.



Analysing the Facts:

After conducting experiments, **the researcher analyzes the results** by measuring **how much contamination is removed by the filters** and **compares the purified water quality to safe drinking standards**.



Reaching Conclusions:

The researcher concludes that **the charcoal and sand filters successfully remove 90% of contaminants** and **can provide safe drinking water for the village**. This solution could either be implemented in the village or generalized for use in other similar locations.

Objectives of Research

- The purpose of research is to **discover answers to questions through the application of scientific procedures.**
- The main **aim of research** is to **find out the truth which is hidden and which has not been discovered as yet.**
- Though each research study has its own specific purpose, we may think of **research objectives as falling into a number of following broad groupings:**

Exploratory or Formulative research studies

To gain familiarity with a phenomenon or to achieve new insights into it.

Example:

Suppose a company wants to explore **why customers are not purchasing a new product**. An exploratory research study could involve **conducting informal interviews with customers to gather insights about their opinions and preferences regarding the product**.

Descriptive research studies

Its all about describing things in detail.

Imagine you want to **study the eating habits of college students**. A descriptive research study would simply gather and present data on things like:

What foods students eat the most?

How often they eat meals?

Where they prefer to eat (home, cafeteria, or outside)?

This type of research **doesn't try to explain why students eat certain foods or how eating affects their health—it just describes their eating habits as they are**.

Diagnostic research studies

The term **diagnostic** refers to the process of identifying, analyzing, or determining the cause of a problem, condition, or situation.

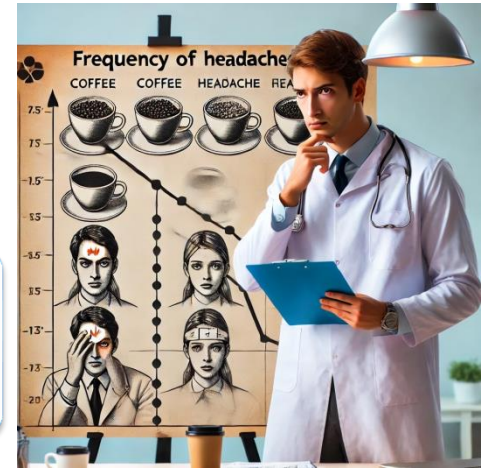
studies are focused on finding out **how often something happens** or **how it is connected to something else**. These studies aim to **identify patterns, causes, or associations between different things**.

Example:

Imagine you're a doctor, and a lot of your patients are complaining about headaches. You want to find out how often these headaches occur and whether they might be linked to something specific, like drinking coffee. So, you do a diagnostic research study.

You ask all your patients how often they drink coffee and how often they get headaches. After collecting the data, you see that people who drink more coffee tend to have more headaches.

This research helps you diagnose that **there's a possible connection between coffee and headaches, or how frequently it happens in your patients**.



Hypothesis-testing research studies

It is a process used in research **to determine whether there is enough evidence to support a particular idea or assumption (called a hypothesis).**

It helps to **test whether one thing (a variable) has an effect on another.**

Steps in Hypothesis testing research studies:

- 1) Formulate the hypothesis
- 2) Collect data
- 3) Analyze the data
- 4) Conclusion

Hypothesis: "Studying for 1 hour every day improves exam scores."

- You create two groups of students: one group studies for 1 hour daily, and the other group doesn't follow a strict schedule.
- After a month, you compare their exam scores.
- If the group that studied regularly scores higher, your hypothesis is supported.
- If there is no difference in scores, the hypothesis might be rejected.

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The **objectives of the engineering research** is:

- 1) The **goal of engineering research** is to **solve important new problems**, and the **outcome is unknown at the start** but **must be new by the end**.
- 2) **Research objectives** can **sometimes be complicated**, but **knowing how and where to find information helps in solving engineering problems**, both academically and professionally.
- 3) **Not following engineering guidelines, standards, or best practices** can lead to **serious failures**, so **accurate research and clear communication** are **crucial for making decisions**.
- 4) The main goal of research is to **use scientific methods to find answers to unanswered questions**. Each study may **require a specific approach suited to the problem**.

5) Engineering research should **aim to develop new theoretical or practical knowledge, not just focus on getting the expected results.**

6) Even if the **desired result isn't achieved, understanding why it didn't work** is still a valuable contribution to solving the problem.

Motivation in Engineering research

1) Intrinsic motivation:

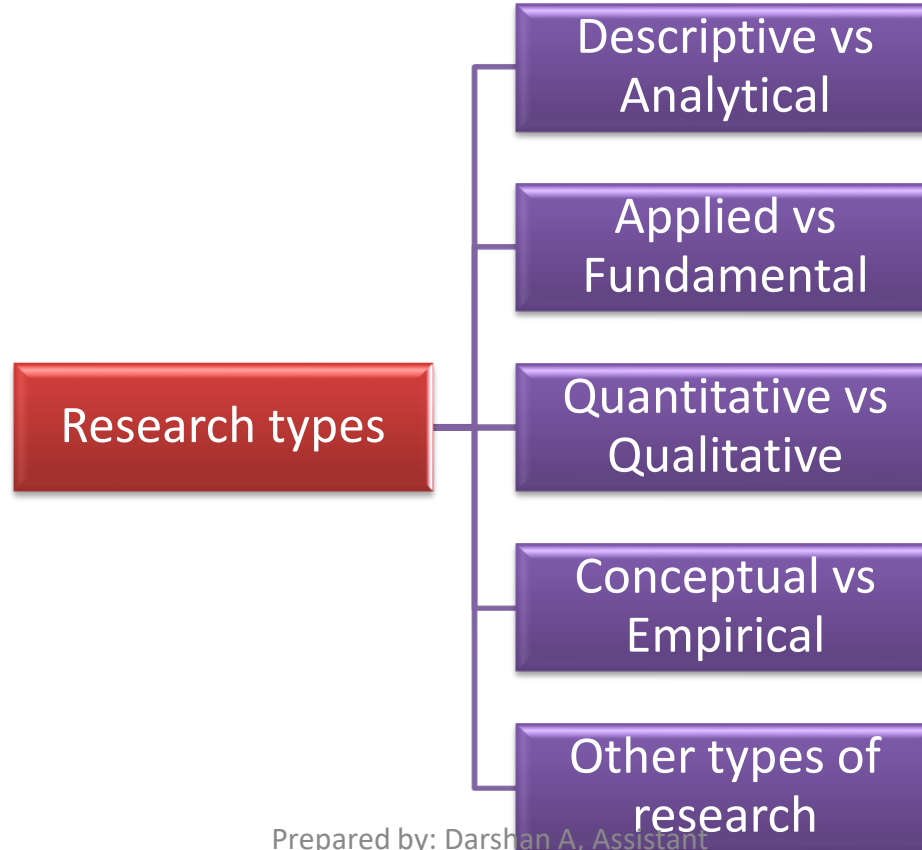
- Studies have shown that intrinsic motivations like **interest, challenge, learning, meaning, purpose**, are linked to strong creative performance;

2) Extrinsic motivation:

- Extrinsic motivating factors like **rewards for good work** include **money, fame, awards, praise, and status** are very strong motivators, **but may block creativity**.
- For example: Research outcome may enable **obtaining a patent** which is a **good way to become rich and famous**.

- ❑ Influences from others like **competition, collaboration, commitment, and encouragement** are also motivating factors in research.
- ❑ **Personal motivation in solving unsolved problems, intellectual joy, service to community, and respectability** are all driving factors.
- ❑ **Wanting to do better than what has been achieved in the world.**
- ❑ Several other factors like **government directives, funding opportunities in certain areas, and terms of employment,** can **motivate people to get involved in engineering research.**

Types of research



1) Descriptive vs. Analytical research

- Descriptive research is a type of research that **aims to accurately describe characteristics, behaviors, or phenomena as they exist in their natural state.**
- It focuses on providing a comprehensive picture of the subject under study without manipulating it.
- It **captures "what is" rather than exploring "why" or "how" things happen.**
- Descriptive research can use **both quantitative data (numerical information) and qualitative data (descriptive information).**
- It **focuses on documenting facts, patterns, or trends without drawing conclusions about cause and effect.**
- **Surveys, observations, and case studies** are common methods used in descriptive research. These methods help gather detailed information about the subject being studied.

Example:

Topic: Studying the Demographics of Smartphone Users

- **Research Question:** What is the age distribution of smartphone users in a particular city?
- **Method:** A survey is conducted to collect data on the age, gender, and occupation of smartphone users.
- **Outcome:** The research describes the demographics, such as "40% of users are aged between 18-24," "30% are aged between 25-34," etc.
- **Conclusion:** This study describes the current distribution of smartphone users without explaining why this distribution exists.



- Analytical research is used to analyze or explain why something happens or how it works. It goes beyond mere description and attempts to understand the underlying reasons or mechanisms.
- It involves delving deeper to understand the underlying reasons, mechanisms, relationships, and cause-and-effect connections between variables. It's about not just observing "what" is happening, but explaining "why" it is happening and "how" the elements are interconnected.
- Analytical research, especially in experimental settings, often involves comparing control groups (not exposed to the experimental variable) with experimental groups (exposed to the variable) to understand the effects of the variable.
- The goal is not just to collect data but to interpret it, providing insights into why certain patterns exist or what the implications of the findings are.
- Analytical research is used in Business and Economics, Health and Medicine, Social Sciences.
- Examples:
 - 1) Impact of drug on health outcomes.
 - 2) Effect of marketing strategies on sales.
 - 3) Analyzing factors influencing academic success. Etc.

Impact of a Drug on Health Outcomes:

- **Hypothesis:** A new drug reduces blood pressure more effectively than the current standard treatment.
- **Method:** Conduct a clinical trial with two groups—one receiving the new drug and the other receiving the standard treatment.
- **Analysis:** Use statistical tests to compare the outcomes (blood pressure levels) between the two groups.
- **Conclusion:** Determine whether the new drug has a statistically significant effect on lowering blood pressure compared to the standard treatment.



Aspect	Descriptive Research	Analytical Research
Purpose	To describe characteristics or phenomena.	To analyze and explain the "why" and "how."
Question	Answers "What is happening?"	Answers "Why is it happening?" and "How?"
Methodology	Surveys, observations, descriptive statistics.	Experiments, statistical analysis, hypothesis testing.
Example Topic	Demographics of smartphone users.	Impact of social media on academic performance.
Outcome	Provides data and descriptive insights.	Provides explanations, relationships, and causality.

2) Fundamental vs Applied

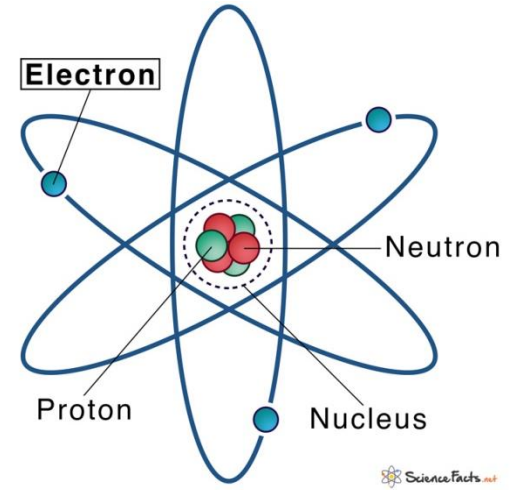
- **Fundamental Research** (also called **Basic Research** or **Pure Research**) is a type of research aimed at expanding knowledge and understanding fundamental principles without a focus on practical applications.
- It seeks to answer questions like "Why does this happen?" or "How does this work?" and helps build a foundation for future discoveries and applied research.
- Purpose of Fundamental Research: To gain a deeper understanding of natural phenomena or basic principles.
- Nature of Fundamental Research: Theoretical, focusing on concepts and ideas rather than practical use.
- Outcome of fundamental Research: Contributes to the overall knowledge base and may lead to new theories or models.

Example:

Research Topic: Studying the properties of electrons.

Description: Scientists investigate how electrons behave under different conditions, such as high speeds or in the presence of magnetic fields. They aren't looking to create a new device but to understand the basic properties of electrons.

Outcome: This research deepens our understanding of atomic structure, which could later be applied to develop new technologies like faster computers or better electronic devices.



Discovery of Electricity: Early research on electricity by scientists like Benjamin Franklin and Michael Faraday was purely fundamental. They studied how electrical charges worked without knowing it would lead to inventions like light bulbs, electric motors, and computers.

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- **Applied Research** is a type of research **aimed at solving specific, practical problems**. Unlike **basic research**, which is **focused on gaining new knowledge without immediate real-world applications**, applied research is used to find solutions that can be directly implemented.
- The primary aim is to **solve a problem or improve processes, products, or technology**.
- Results are intended to have a direct effect on society, businesses, or technology.

Example:

Objective: Develop a new pain-relief medicine.

Problem: Current medicines may have side effects or limited effectiveness.

Research Process:

- Test various chemical formulas.
- Analyze their effectiveness in relieving pain.
- Evaluate safety and side effects.

Outcome: Identify the best formula that is both effective and safe.

Application: Use the research findings to create a new, improved pain-relief product for consumers.

3) Quantitative versus Qualitative

Quantitative Research

Definition: Quantitative research deals with numbers and statistics. It helps us measure and compare things to find patterns or make predictions.

Example:

- **Student Grades:** A teacher collects test scores from 100 students to see the average score and find out how many students passed or failed.

Qualitative Research

Definition: Qualitative research focuses on understanding feelings, thoughts, and experiences. It involves talking to people or observing them to get deeper insights.

Example:

- **Student Experiences:** A teacher interviews 10 students to understand how they feel about online classes and what challenges they face.

Example: Quantitative research

Customer Satisfaction Survey:

1. **Scenario:** A company wants to know how satisfied its customers are with a new product.
2. **Method:** They create a survey with questions like:
 - "How would you rate the product on a scale of 1 to 5?"
 - "How likely are you to recommend this product to others? (1 to 10 scale)"
3. **Data Collection:** The survey is sent to 500 customers, and responses are collected in numbers.
4. **Analysis:**
 - Calculate the average satisfaction score.
 - Find out what percentage of customers rated the product 4 or 5 out of 5.
5. **Outcome:** The company learns that 80% of customers gave a high satisfaction score (4 or 5).
This helps them understand that most customers are happy with the product.

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Example: Quantitative research

Imagine a school wants to know how students feel about their new online classes. They interview 15 students and ask open-ended questions like:

- “How do you feel about learning from home?”
- “What challenges are you facing with online classes?”
- “What do you like or dislike about the new system?”

The students share their personal experiences, concerns, and suggestions. The school then analyzes these responses to identify common themes, like technical issues, lack of interaction, or benefits such as flexible timing. This helps the school understand students' perspectives and make improvements.

4) Conceptual versus Empirical

- Conceptual research **focuses on abstract ideas and theories**. It involves **exploring existing concepts, theories, or frameworks without necessarily collecting new data**. Researchers **aim to develop or refine theories, definitions, or concepts**.
- Example: **Studying the Benefits of Meditation**

Conceptual Research: A researcher reviews existing theories and ideas about meditation. They **explore the psychological and spiritual concepts behind meditation** and **propose a new theory about how meditation can reduce stress**. This research is based on thinking and developing ideas, **without collecting new data**.

- **Empirical research**, on the other hand, is **based on observation or experiments**. It involves **collecting data through methods like surveys, experiments, or case studies to test hypotheses or theories**.

Example: **Studying the Benefits of Meditation**

Another researcher conducts an experiment where a **group of people practices meditation daily for a month**. They **measure stress levels before and after the experiment using surveys and health data**. This research involves **collecting real-world data** to test if meditation actually reduces stress.

“In this example, the conceptual research focuses on **developing ideas**, while empirical research focuses on **testing** those ideas with data.”

FINDING AND SOLVING A WORTHWHILE PROBLEM

1) Choosing a Problem

- A researcher might pick a problem **suggested by a supervisor** or **identify a new one by reviewing research papers**. It's essential to develop skills for selecting a problem, but these skills are often not taught directly.

2) Initial steps

- Before starting, it helps **to have an idea or "spark."** Sometimes, **inspiration comes from listening to presentations or noticing developments in other fields that relate to your area of research.**

3) Worthwhile problem

- Unexpected or surprising.
- Something long-anticipated by the research community.
- A major simplification of a theory.
- A new result that opens a new field.
- A method that improves existing techniques or has practical use.
- A result that stops further work in a certain area.

FINDING AND SOLVING A WORTHWHILE PROBLEM

4) Effort and Acceptance

- The **researcher needs to believe the problem is worth solving**, as this will drive the best efforts. Not all problems need to be groundbreaking; **solving smaller problems effectively can also lead to significant progress.**

5) Challenges

- **Hard problems might take a long time to solve, and not everyone succeeds in solving them.** Still, trying to solve hard problems is important.

6) Partial Success

- Even **if the main problem isn't solved**, **partial results can be useful and contribute to the dissertation.**

Polya's Problem-Solving Steps:

- Understand the problem deeply.
- Start somewhere and explore strategies, even if they're simpler versions of the problem.
- Execute your plan and be ready to start over if it fails.
- Reflect on the process to improve your problem-solving skills for the future.

Example of Ethics

A Clinical Trial for a New Drug

- Developing a new drug to treat chronic illness and it is now in human trial phase.

1) Informed consent:

- Volunteer is informed about the purpose of the study, potential risks, benefits, and alternatives.

Ethical Principle: Participants must give **voluntary consent** without any pressure. They should be aware of what the study entails.

2) Minimizing Harm:

During the preclinical phase, the drug was tested on animals and in laboratories to minimize harm in human trials. Researchers ensured that only participants in stable condition were recruited to minimize risk.

Ethical Principle: Researchers must ensure they avoid **unnecessary harm** to participants, protecting their well-being throughout the process.

3) Risk-Benefit Analysis:

The researchers assess the risks of side effects like nausea or headaches versus the potential benefit of reducing the chronic illness's severity.

Ethical Principle: The risk to participants must be **proportional to the potential benefits** of the research.

4) Freedom to Withdraw:

Ethical Principle: Participants must have the **freedom to withdraw** from the study at any time without repercussions.

5) Privacy and Confidentiality:

The personal health information of participants is kept confidential. Results are anonymized in reports, and only authorized personnel have access to individual data. **Ethical Principle:** Researchers must protect the **privacy and confidentiality** of participants.

6) Fair Participant Selection:

The trial includes a diverse group of participants of different ages, genders, and ethnicities to ensure that the drug's effects are well understood across different populations.

Ethical Principle: There must be **fair and equitable selection** of participants, avoiding exploitation of vulnerable groups.

7) Independent Review

Before the trial begins, an **Institutional Review Board (IRB)**, an independent ethics committee, reviews the study's design to ensure it adheres to ethical standards and safeguards the rights of participants.

Ethical Principle: An **independent review** helps to avoid conflicts of interest and ensures the study meets ethical standards.

ETHICS IN ENGINEERING RESEARCH

- **Ethics** is like a **set of rules** that helps us to know **what is good and what is not**.
- Most **people begin learning ethical norms during childhood, but their understanding of morality evolves as they grow**. While basic ethical principles are widely recognized, **how these norms are interpreted and applied can differ across individuals and cultures** due to varying perspectives and experiences.

An example of this concept is **honesty**. Most people are taught from a young age that **lying is wrong** (a basic ethical principle). However, how honesty is interpreted and applied can vary across cultures and situations.

For instance:

- In some cultures, **telling a white lie** (a small, harmless lie) to avoid hurting someone's feelings is considered acceptable or even polite.
- In other cultures, strict honesty, even if it might cause discomfort or offense, is seen as the right thing to do.

Both groups value honesty, but their **application of the principle differs based on cultural perspectives and experiences**.

- **Ethical principles** help **guide us in deciding what is right or wrong**.
- These principles can be used to **evaluate, suggest, or understand laws**.
- While ethics and laws are different— **ethics are moral guidelines**, and **laws are official rules** — **laws often reflect ethical values** because **they represent what society agrees is important**.
- For example, laws against stealing exist because stealing goes against shared ethical beliefs about fairness and respect for others' property.
- **International norms** for the **ethical conduct of research** have been there **since the adoption of the Nuremberg Code in 1947**.
- According to **Whitbeck**, the concerns about **giving credit for research** go back to the **founding of the British Royal Society (BRS) in the 17th century**. This **organization aimed to improve the methods and practices of modern science**. A key change they introduced was that **credit for research findings went to the person who first submitted their work for publication, rather than determining who originally discovered the idea**. This shift **impacted how research results were shared and recognized in the scientific community**.

The **Nuremberg Code (1947)** outlines 10 ethical principles for human experimentation, established after World War II. Key points include:

1. **Voluntary consent** of participants is essential.
2. Research must benefit society and be based on prior knowledge.
3. Harm to participants must be avoided, and the risk should not outweigh the benefits.
4. Participants should have the right to withdraw anytime.
5. Qualified researchers must conduct the experiments, and studies should stop if they pose danger.

These principles ensure the protection, rights, and safety of human subjects in research.

- **Whitbeck** highlights two important questions about authorship in research:
 1. **Who should be an author?:** This question asks who deserves credit for the research. It's **important to decide which contributors have played a significant role in the research process and should be recognized as authors.**
 2. **What is the correct order of authors?:** This question addresses how to list the authors. The **order can indicate the level of contribution, with the first author typically being the one who contributed the most, followed by others based on their level of involvement.**
- These **questions are crucial for ensuring fairness and clarity in acknowledging contributions to research work.**

- **Co-authorship in an Interconnected World:** In today's world, where **researchers often collaborate**, the issue of co-authorship is very important. Sometimes, **people who play a significant role in conducting research may not be involved in writing the final report or paper**, yet they **still deserve recognition for their contributions**.
- **Codes for Research Ethics:** Many governments and universities have established guidelines for research ethics. These **guidelines help ensure that research is conducted fairly and responsibly**. However, people often confuse "research ethics" with "responsible conduct of research."
- **Difference Between Research Ethics and Responsible Conduct:**
Research Ethics: Research ethics is about thinking carefully about **how the results of research are used**.

For example, when **research leads to new knowledge, ethics makes sure that this knowledge is used in a way that doesn't harm people or society**. It looks at whether the **outcomes are applied in a fair and responsible way**.

Responsible Conduct of Research: This refers to the **proper way to conduct research, including following ethical guidelines and best practices during the research process itself**.

ETHICS IN ENGINEERING RESEARCH PRACTICE

1) Technological advancements create ethical concerns like privacy and surveillance.

Example: A company develops facial recognition technology for security purposes, but it can be misused for mass surveillance, violating individuals' privacy. Ethical concerns arise about how and where this technology should be used.

2) Engineering researchers must make ethical choices, especially when their work impacts people

Example: A researcher working on AI for medical diagnosis must ensure that the system does not exhibit bias against certain populations, as it could lead to incorrect treatment for some groups.

3) We have access to vast data, but not everything that can be done with it should be done.

Example: Social media platforms collect massive amounts of user data. While this data can be analyzed to improve user experience, it can also be sold to third parties without user consent, which is unethical.

4) Ethics guide engineers in deciding what is right and wrong in their work

Example: An engineer designing self-driving cars must decide how the car should react in a situation where it must choose between hitting a pedestrian or risking harm to the driver. Ethical guidelines help shape these decisions.

5. Researchers influence technology by setting ethical standards and making choices during design and development:

Example: A team developing a new type of drone could prioritize privacy by ensuring the drone is only capable of capturing data in public areas, not private spaces like homes.

6. It's important to consider safety and minimize risks in research outcomes to avoid harm:

Example: When developing a new type of battery, a researcher must ensure it is safe by including protective layers to prevent it from overheating and causing fires, which can lead to dangerous accidents

TYPES OF RESEARCH MISCONDUCT

1. Fabrication (Creating fake data or results):

- Fabrication refers to making up or inventing data, results, or experiments that were never conducted. This is often done under time pressure, or when researchers believe they know the outcome but don't want to wait for actual results.
- Researchers might feel pressure from supervisors, customers, or deadlines to present results even though the experiments or data collection haven't been completed. This falsified information is then presented as legitimate.

2. Falsification (Altering data to fit desired outcomes):

- Falsification involves manipulating research data, processes, or changing the results so they support a hypothesis that might otherwise be unsupported by the original data.
- This could include modifying graphs, altering experimental setups, or omitting outlier data points to make the data look more convincing or to align with expected outcomes.

Fabrication = Making up fake data/results.

Falsification = Altering real data to mislead.

3. Plagiarism (Using someone else's work without proper credit):

- Plagiarism occurs when a researcher uses another person's ideas, words, data, or figures without proper acknowledgment. It can include copying large parts of text or data, or even reusing one's own previously published work (self-plagiarism) without citation.
- The availability of research papers online makes it easier to copy work, but also makes it easier to detect plagiarism through software tools. Even if you rephrase someone else's work, you still need to give proper credit.

4. Other Aspects of Research Misconduct

- **Simultaneous Submission:** Submitting the same research paper to multiple journals or conferences at the same time without informing the publishers is unethical because it wastes resources and undermines the peer review process.
- **Failure to Report Mistakes:** Mistakes in published work often go unreported unless another researcher finds them and corrects them in subsequent research. This can delay the advancement of the field.

Impact of Misconduct:

- Misconduct like fabrication, falsification, and plagiarism hampers scientific progress by introducing false data into the literature, damaging the reputation of researchers and their institutions, wasting time and resources, and delaying technological advancements.

ETHICAL ISSUES RELATED TO AUTHORSHIP

1. Academic Authorship:

- Authorship is vital for communicating research, establishing priority for discoveries, and gaining recognition. It's a key factor in evaluating researchers for employment, promotions, and awards.
- Authorship requires taking full responsibility for the content of the work.

2. Types of Research Contribution Credits:

- **Authorship:** For significant contributors involved in research design, data interpretation, or writing.
- **Citation:** Referring to previously published work.
- **Acknowledgment:** Recognizing minor contributions that do not warrant authorship.

3. Ethical Issues:

- **Guest or Gift Authorship:** Listing someone as a co-author who made little or no contribution to the research.
 - **Example:** A researcher includes a colleague's name as a co-author despite them not participating in the work, to boost the colleague's career.
- **Career-Boost Authorship:** Adding junior faculty or students as co-authors to help their career, even if they didn't contribute significantly.
 - **Example:** A professor lists a student as a co-author to improve their resume, although the student contributed minimally.
- **Career-Preservation Authorship:** Adding administrators (like deans or heads of departments) as co-authors to maintain good relationships.
 - **Example:** A researcher includes their department head as a co-author to gain favor, even though the head had no involvement in the research.

- **Ghost Co-authorship:** When someone who contributed is not listed as an author due to conflicts of interest or other reasons.
 - **Example:** A significant contributor is left out from the author list because of internal organizational conflicts.
- **Reciprocal Authorship:** Researchers agree to list each other as co-authors as a favor, without substantial collaboration.
 - **Example:** Two researchers add each other's names to their papers as co-authors, even though they only briefly reviewed the other's work.

4. Double Submission:

- Submitting the same paper to two different journals or conferences simultaneously, which violates ethical publication practices.
 - **Example:** A researcher submits their article to two journals at the same time, hoping for quicker publication, which is unethical as journals demand original, unpublished work.